Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 · Energy storage devices, hydrogen energy, and sensors · CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells

1 CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells

Study focus: CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000118 · EV0000119

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll continue with module 5 on energy storage devices that is on fuel cells.

Key details

- This fuel cell is also known as proton exchange membrane fuel cells, thus telling us that the polymer which we are going to use is capable of exchanging protons.
- Why is methanol fuel cell so popular these days.
- If you remember in the hydrogen oxygen fuel cell I mentioned that storage and transport of hydrogen is going to be difficult because it is in the form of a gas.
- But methanol being a liquid.
- We can easily transport and store the methanol, and hence there is a possibility that we can use it for portable devices.
- This fuel cell has low weight and volume.
- Low weight and volume because we are we are using polymer as the electrolyte and polymers are lightweight.
- So therefore this cell will have very low weight and volume and therefore.
- What is the good characteristic of this fuel cell?
- That it is going to have very high energy density.
- The weight has reduced, therefore the energy density will be very high.
- But there is one problem we face with this fuel cell, that we have to use only noble metal as catalyst, and usually it is platinum which is employed as the catalyst in these cells.

- And one more problem which we face that carbon monoxide.
- If it is present, then it is going to poison the catalyst.
- We are using platinum as the catalyst.
- It is going to poison the catalyst, and therefore pure fuel or oxidant only should be used.
- The electrode is a porous carbon electrode containing platinum.
- Cathode is also a porous carbon electrode containing platinum.
- Just now I mentioned that noble metal only should be used, but then we always have to have a porous electrode.
- because all reactions are going to take place on the electrode.
- The catalyst is present there, the electrolyte is in contact, and also the fuel or the oxidant is going to come in contact with the electrode.
- It is a solution of methanol.
- We never use pure methanol.
- We are going to use a solution of methanol and the oxidant is oxygen gas.
- of methanol and the oxidant is a oxygen is the oxygen gas and the electrolyte as we are studying now it is a proton conducting polymer membrane now any electrolyte for that matter.
- What should be the criteria that it has to be an electronic insulator?
- What does it mean?
- That it is not going to allow the electrons to move through itself that is the electrolyte should be an electronic insulator.
- Yes, we are using a polymer here, so this membrane is an electronic insulator, but at the same time we want the hydrogen ions to move.
- That's why we have chosen a proton exchange membrane.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 05:00–09:59

Excellent conductor of the hydrogen ions.

Key details

- Electrolyte is usually the most common electrolyte used is a fluorocarbon backbone.
- It's similar to Teflon.
- Teflon how does it look?
- It's a CF2, CF2, CF2.
- You have so many units of CF2.
- Which are bound together and that is Teflon.
- Because we know that sulfonic acid has.
- The SO3 minus and H plus it can dissociate in water as SO3 minus and H plus so that this proton now is free to move.
- It becomes like this in the presence of water.
- The H plus ions obviously are going to be able to migrate through when the membrane is hydrated.

- So when because when it is hydrated, you will be able to get the H plus ions to move through the electrolyte.
- The most common one in any protonic stain membrane electrolyte fuel cell or we can call it as a polymer electrolyte fuel cell.
- It's always napion.
- It is a very popular polymer which is used in these fuel cells and that's because it has very high proton conductivity that's what we want the diode should move where very easily through the membrane and it has good mechanical strength and it also has good chemical stability.
- As I said, there's a backbone of CF2, and then that is the Teflon backbone, and then here we see the SO3H groups which have been appended onto this backbone.
- is acquebn which also has a similar structure just that the CF2 branching I mean the CF2 chain has been reduced.
- Here it is a longer chain and here you can see it's a smaller chain.
- Nevertheless Nepheon happens to be the most popular membrane which is being used.
- But when we come to DMFC, what is DMFC?
- Direct methanol fuel cells.
- That is what we are studying now.
- Nafheon is not suitable for such a fuel cell.
- where we are using methanol and it's a diet methanol fuel cell.
- Because along with H plus ion movement naphion is highly permeable to methanol.
- What happens because of that?
- It's allowing the methanol also to move through the membrane and therefore a phenomenon known as methanol crossover happens.
- Here you can see the methanol diffuses through the membrane to the cathode.
- So instead of reacting at the anode undergoing oxidation at the anode we find that methanol just crosses over to the cathode and there it is going to combine with the oxygen which is not what we want because if we want to generate power from the cell then oxidation of methanol has to happen at the anode but.
- But unfortunately, when we use Nafion, it is found that this methanol crossover happens.
- So, therefore, that will result in what you will get lower voltage because less number of electrons are being generated.

Example or application from the lecture

- Let us see.
- Ether ketone the structure is given here.

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 10:00–14:59

Many new membranes are there.

Key details

- Sulfonated polyether ether ketone is used as the membrane.
- We can see the picture here that.
- We have methanol and water which is coming from here.
- And what would be the reaction?
- We are letting in oxygen.
- From the anode to the cathode, and it is going to take up the electrons which are coming through the external circuit and form water.
- exactly what is happening here.
- A weak solution of methanol is fed.
- One reason for a weak solution being used is that methanol crossover is minimum when we use a weaker solution.
- And then the common operating temperatures, as already mentioned, is 60 to 90 degrees centigrade, and the oxidation of methanol on a catalyst layer takes place, and carbon dioxide is.
- What is your electrolyte?
- It is a polymer electrolyte membrane or a proton exchange membrane.
- If the membrane is hydrated the H plus ions will.
- Just hop through the electrolyte and reach the cathode.
- Oxygen reacts with this H plus ions as we saw in the reaction, and then it is going to give out.
- Anode the water is being consumed at the anode because we said a weak solution of methanol is being fed over there.
- Travel through an external circuit.
- Water management in this fuel cell is very important and the temperature has to be maintained 60 to 90 degrees.
- We cannot have a higher temperature, we cannot have a lower temperature.
- Unless it is hydrated you cannot have the H plus ion moving through because only then the SO3H is going to dissociate.
- and water which is produced during the reaction it has to be removed from the cathode very frequently.

Example or application from the lecture

- One example is SPEEK.
- with water reacts with water a solution of methanol we have used and that is going to generate the H plus and we are interested in these H plus ions and electrons are given out plus CO2.
- Being given out as we saw in the reaction, we get carbon dioxide as a product, and at the same time, protons are being generated, and these protons now are going to migrate through the electrolyte.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 15:00–19:59

This is called water management.

Key details

- Now when we have high temperature, what is the problem if we use high temperature what will happen the polymer may get dehydrated and therefore H plus conductivity cannot take place and also the polymer may degrade and crack resulting in short circuiting.
- And another reason for this, another fact which we have to know is that.
- Neither can low temperatures be used because this will result in flooding the cell because water will not be removed if you keep it at a very low temperature.
- Water will not be removed and this will result in flooding of the cell and that reduces the efficiency of the cell.
- And also because the temperature is very low, we will require a higher loading of catalyst.
- We are using platinum as the catalyst.
- we have we it will be very expensive.
- Neither can we use a high temperature, neither can we use a low temperature.
- What are the advantages of this fuel cell?
- So as mentioned it is it has a very high energy density because of the low weight of the polymer.
- And then it has a long life because we are using work making it work at a very low temperature and therefore the with low temperature it will start very quickly.
- first of all, and it also results in less wear and tear of the system components and therefore a better durability.
- No free corrosive liquid is being used in the cell because the electrolyte is a polymer.
- Therefore, there is no free corrosive liquid which is being used in the cell.
- Noble metal catalysts have to be used and the electrolyte is also expensive.
- We are using special electrolytes, special proton exchange membranes have to be used.
- And if we have CO then carbon monoxide if at all present it is going to poison the platinum catalyst and hence pure reactant should be.
- And management of carbon dioxide, which is created at the anode and water management pose problems.
- As I mentioned that water shouldn't be present in a right amount, when in the right amount and it should not be in excess.
- just now we discussed water management.
- In the next class, we are going to look at another fuel cell that is a solid oxide fuel cell where we are going to use carbon monoxide as the fuel.
- fuel cell using solid oxide as the electrolyte.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9968; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

C Revision and retrieval practice